Block

File: Block.kicad_sch

Alimentacion

File: Alimentacion.kicad_sch

Power

File: Power.kicad_sch

ESP32

File: ESP32S3.kicad_sch

Expansion

File: Expansion.kicad_sch

XMOS

File: XMOS.kicad_sch

DAC

File: DAC.kicad_sch

LED & Connector

File: LED & Connector.kicad_sch

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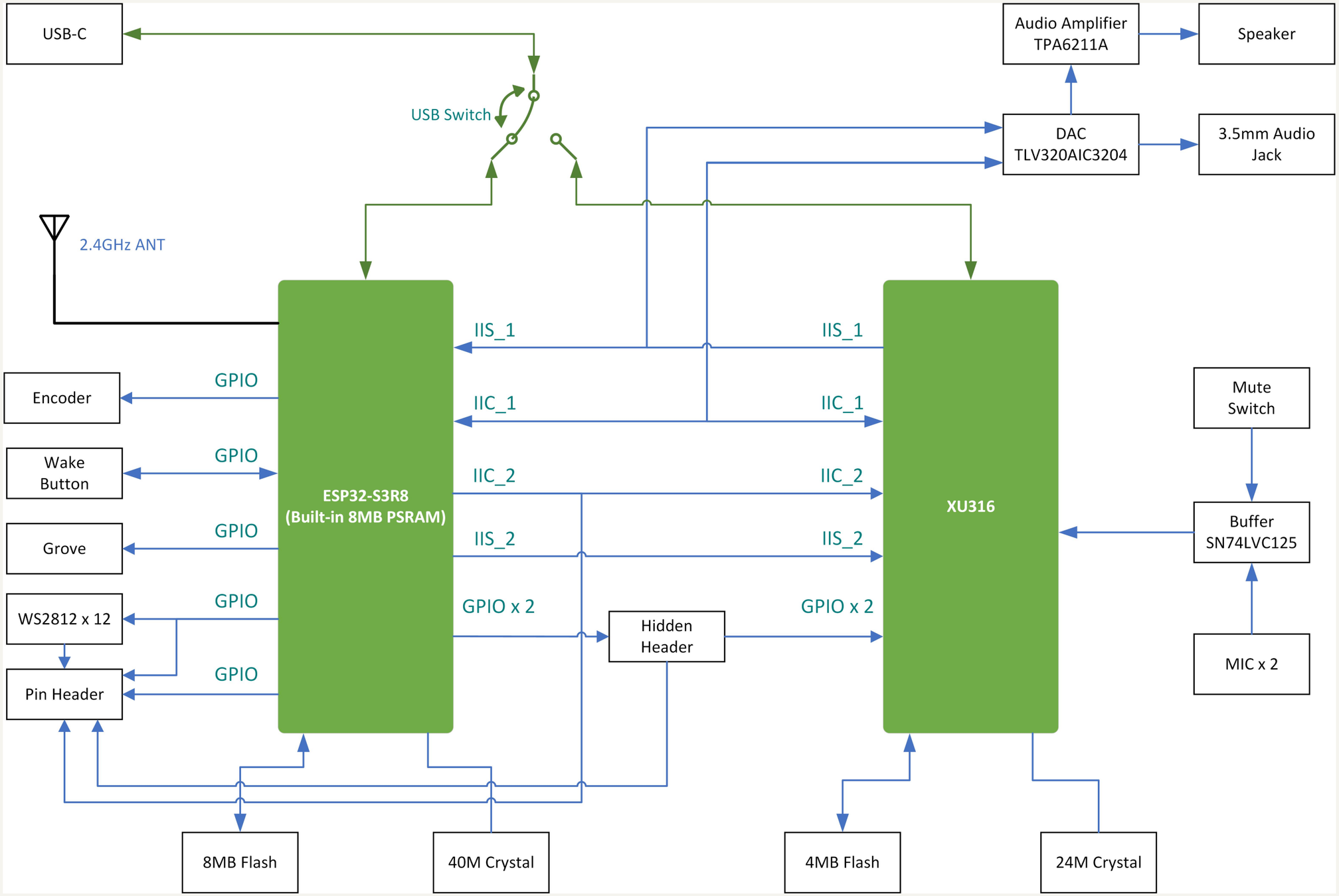
NEO v2 (techmiguel), diseño derivado de Home Assistant Voice Preview Edition (Nabu Casa, CERN-OHL-P v2).
Hojas de Nabu Casa: Block, ESP32, XMOS, DAC, Power, LED & Connector (cambios mínimos documentados en
hardware/neo-v2/docs/fase3-esquematico.md). Hojas propias de NEO v2: Alimentacion y Expansion.

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Sheet: /
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Size: A3	Date:	**Rev:**
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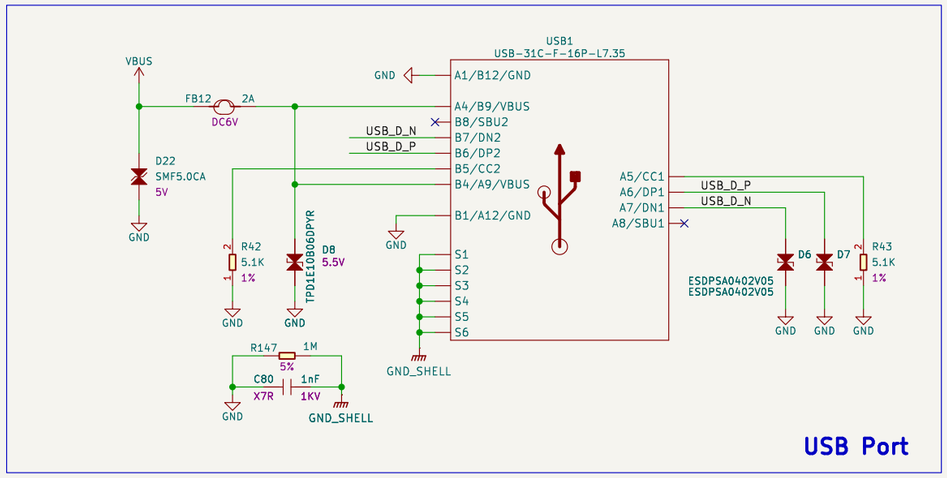


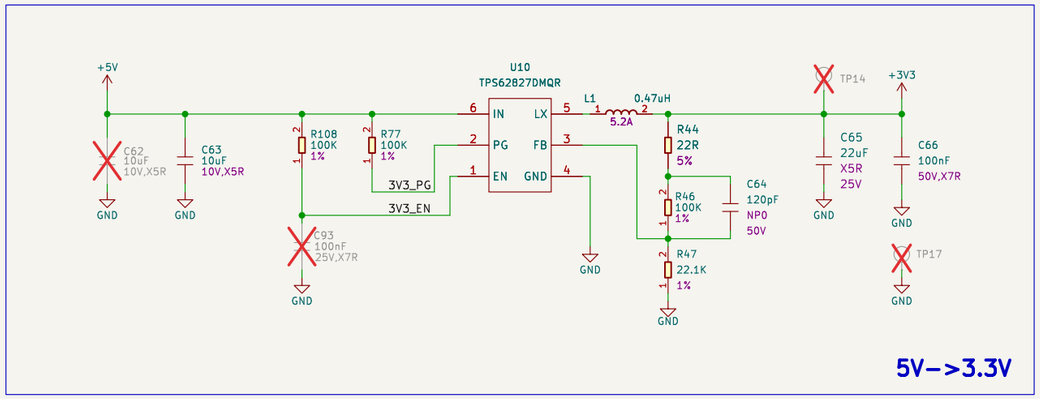
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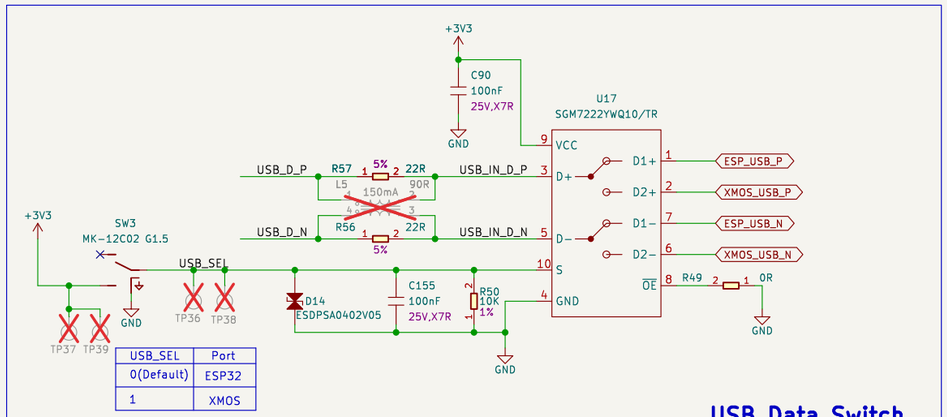
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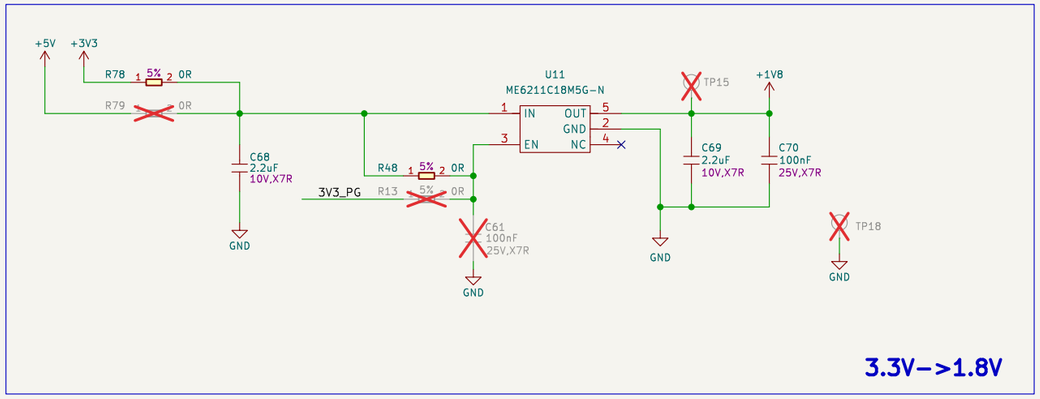
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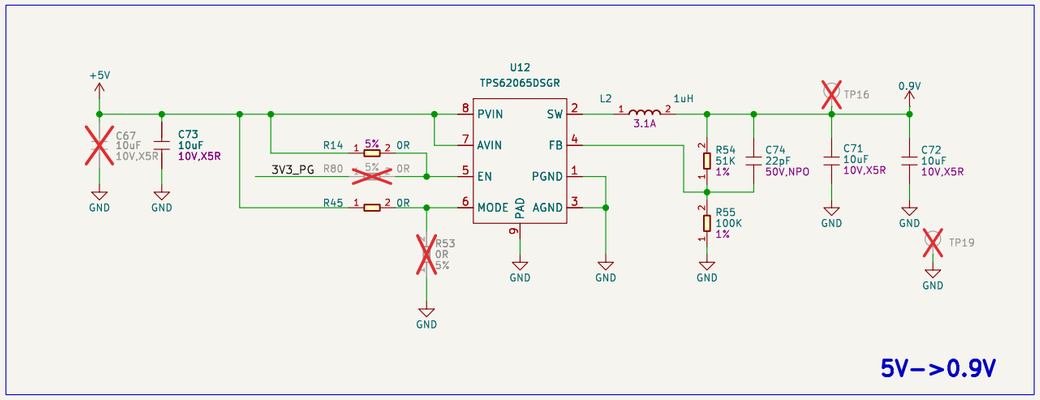
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KiCad E.D.A. 9.0.9		Id: 3/9

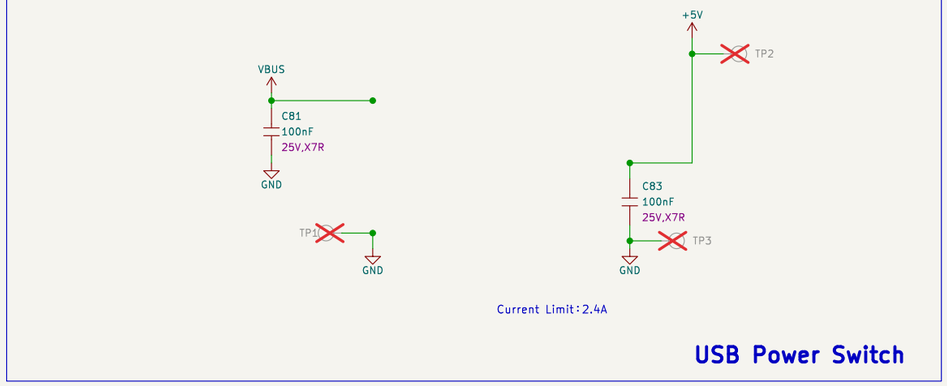










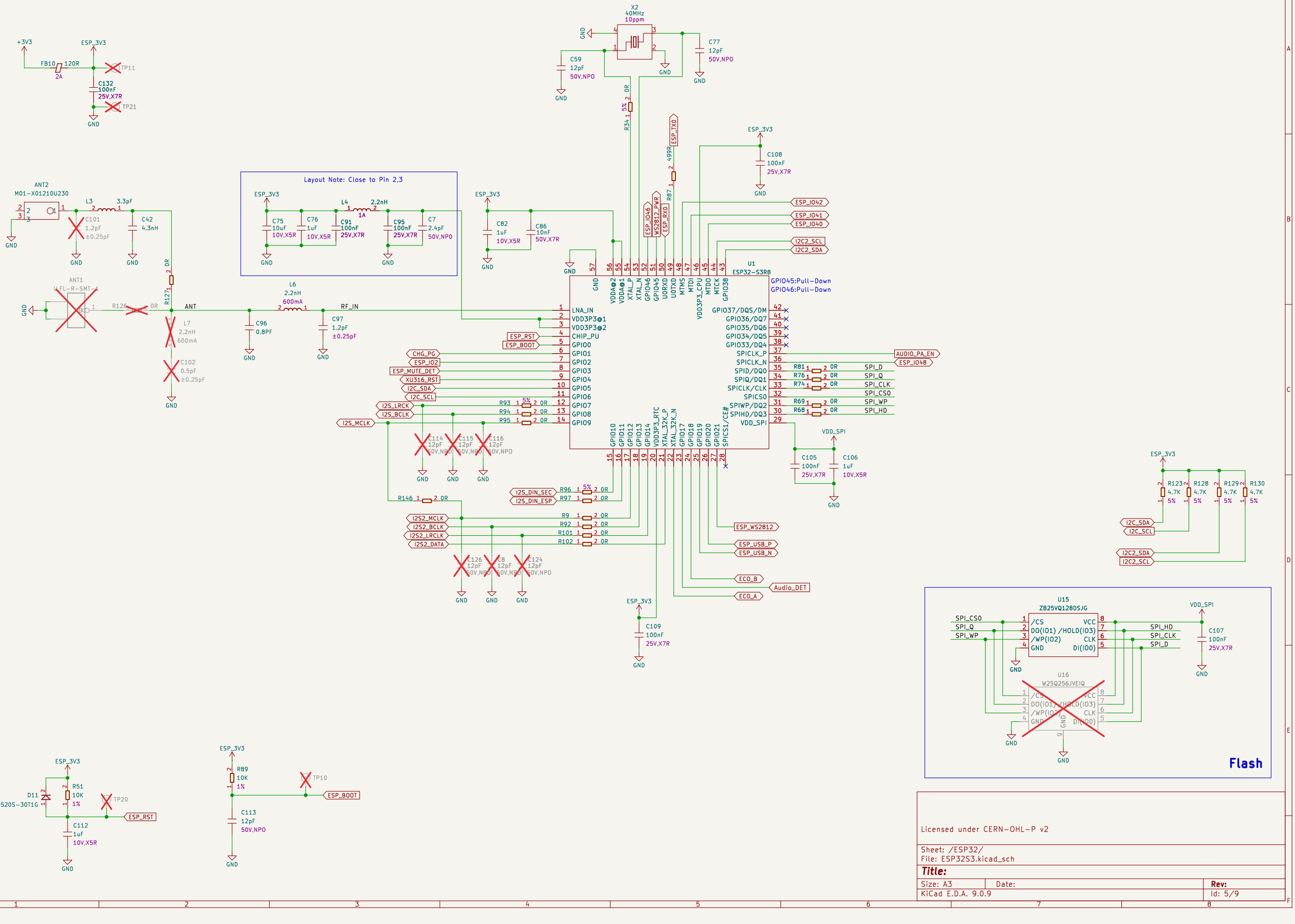


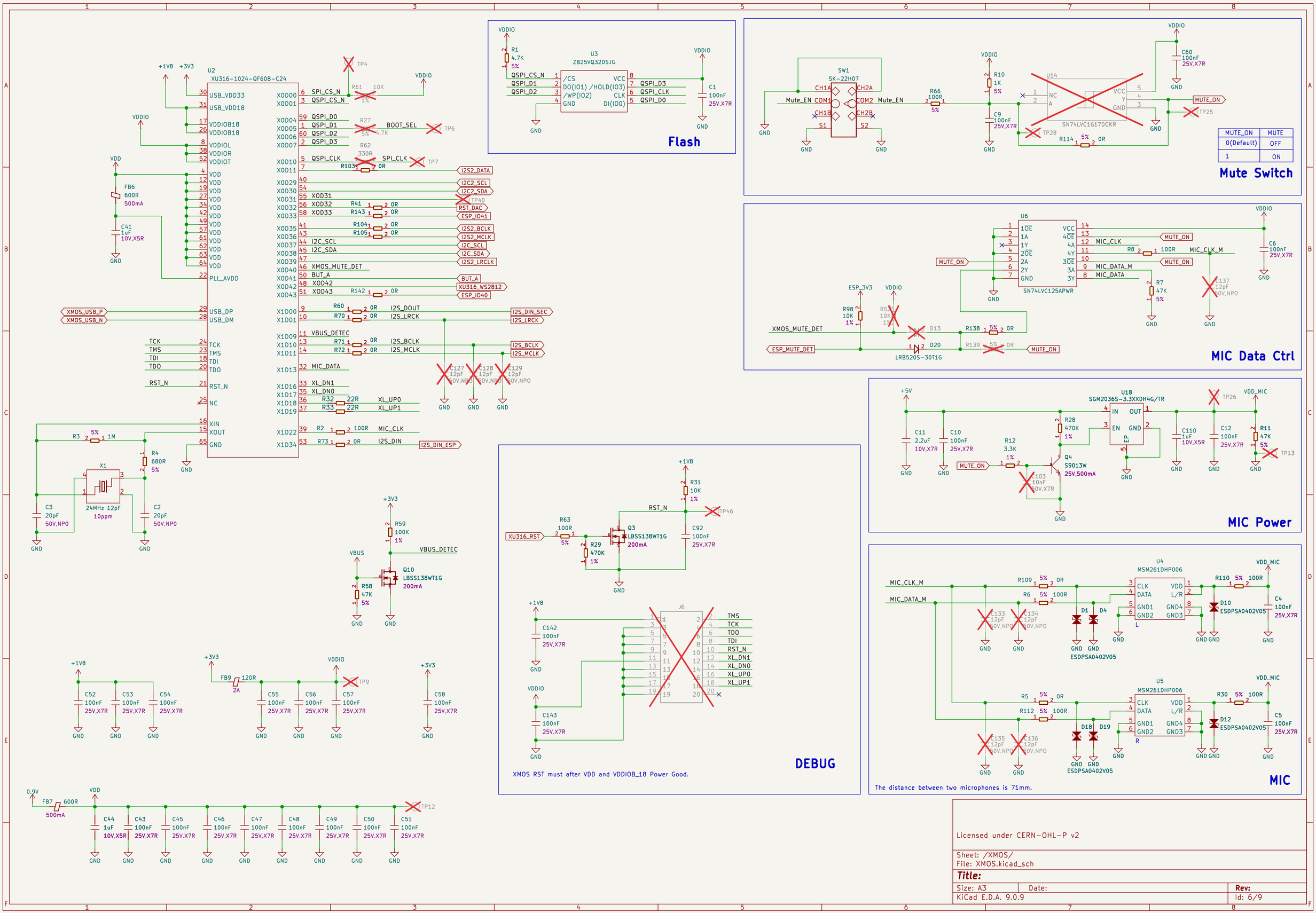
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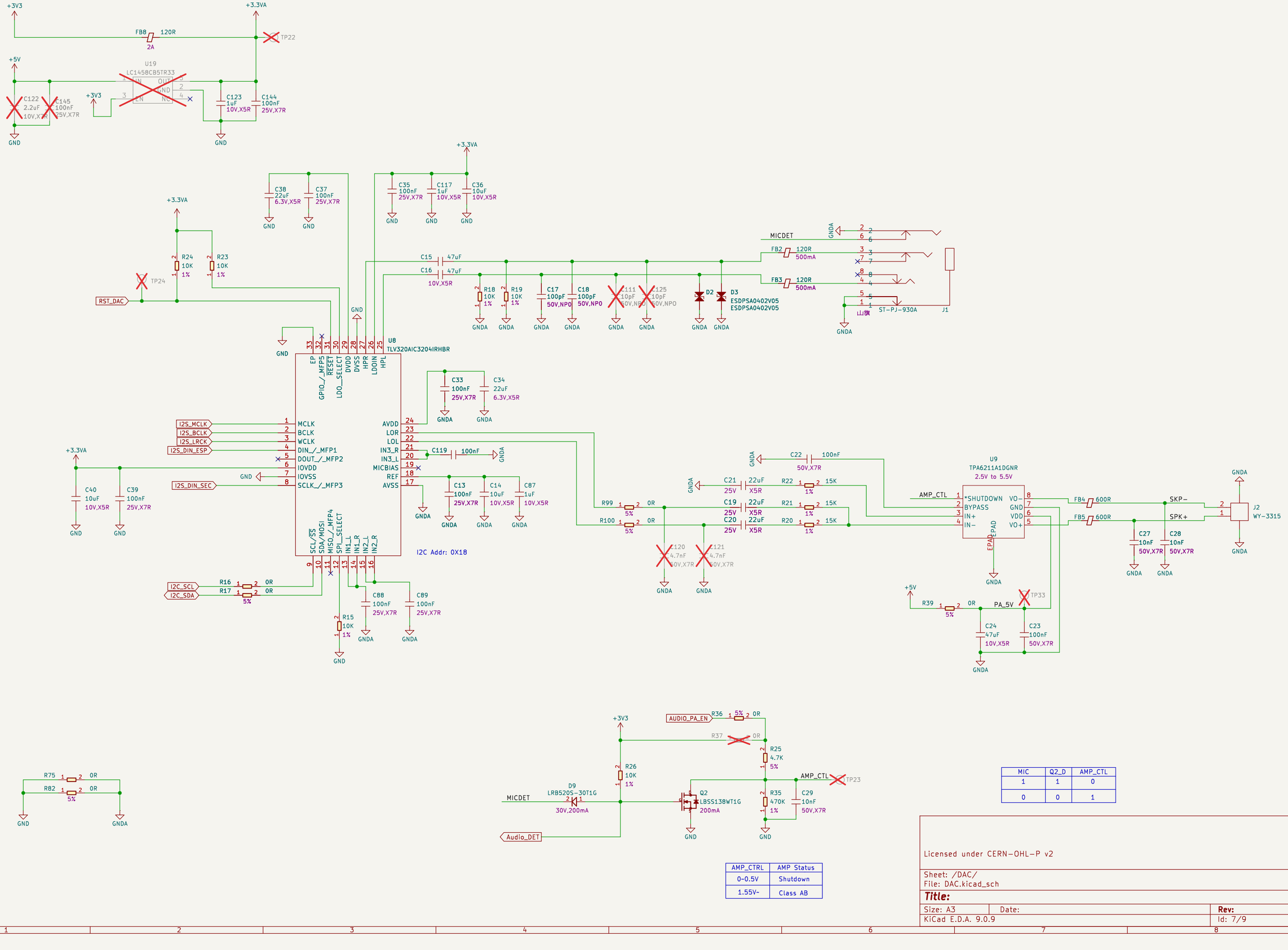
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File: Power.kicad_sch

Title:

Size: A3	Date:	**Rev:**
KiCad E.D.A. 9.0.9		Id: 4/9


+3V3
ESP_3V3
FB10
120R
2A
TP11
C132
100nF
25V,X7R
TP21
GND
X2
40MHz
10ppm
C59
12pF
50V,NPO
C77
12pF
50V,NPO
R34
2 0R
5%
ESP_TXO
499R
R87
C108
100nF
25V,X7R
ESP_IO42
ESP_IO41
ESP_IO40
I2C2_SCL
I2C2_SDA
ESP_IO46
WS2812_PWR
ESP_RXO
ANT2
M01-X01210U230
L3
3.3pF
C101
1.2pF
±0.25pF
C42
4.3nH
Layout Note: Close to Pin 2,3
C75
10uF
10V,X5R
C76
1uF
10V,X5R
L4
2.2nH
1A
C91
100nF
25V,X7R
C95
100nF
25V,X7R
C7
2.4pF
50V,NPO
C82
1uF
10V,X5R
C86
10nF
50V,X7R
U1
ESP32-S3R8
GPIO45:Pull-Down
GPIO46:Pull-Down
ANT1
U.FL-R-SMT-1
R126
0R
R127
2 0R
ANT
L6
2.2nH
600mA
RF_IN
C96
0.8PF
C97
1.2pF
±0.25pF
L7
2.2nH
600mA
C102
0.5pF
±0.25pF
LNA_IN
VDD3P3@1
VDD3P3@2
CHIP_PU
GPIO0
GPIO1
GPIO2
GPIO3
GPIO4
GPIO5
GPIO6
GPIO7
GPIO8
GPIO9
ESP_RST
ESP_BOOT
CHG_PG
ESP_IO2
ESP_MUTE_DET
XU316_RST
I2C_SDA
I2C_SCL
I2S_LRCK
I2S_BCLK
I2S_MCLK
R93 5% 2 0R
R94 2 0R
R95 2 0R
C114 12pF 50V,NPO
C115 12pF 50V,NPO
C116 12pF 50V,NPO
I2S_DIN_SEC
I2S_DIN_ESP
R96 5% 2 0R
R97 2 0R
R146 2 0R
I2S2_MCLK
I2S2_BCLK
I2S2_LRCLK
I2S2_DATA
R9 2 0R
R92 2 0R
R101 2 0R
R102 2 0R
C126 12pF 50V,NPO
C8 12pF 50V,NPO
C124 12pF 50V,NPO
ESP_WS2812
ESP_USB_P
ESP_USB_N
ECO_B
ECO_A
Audio_DET
GPIO37/DQS/DM
GPIO36/DQ7
GPIO35/DQ6
GPIO34/DQ5
GPIO33/DQ4
SPICLK_P
SPICLK_N
SPID/DQ0
SPIQ/DQ1
SPICLK/CLK
SPICS0
SPIWP/DQ2
SPIHD/DQ3
VDD_SPI
AUDIO_PA_EN
ESP_IO48
R81 2 0R SPI_D
R76 2 0R SPI_Q
R74 2 0R SPI_CLK
SPI_CS0
R69 2 0R SPI_WP
R68 2 0R SPI_HD
C105
100nF
25V,X7R
C106
1uF
10V,X5R
C109
100nF
25V,X7R
R123
4.7K
5%
R128
4.7K
5%
R129
4.7K
5%
R130
4.7K
5%
U15
ZB25VQ128DSJG
/CS
DO(IO1) /HOLD(IO3)
/WP(IO2)
GND
VCC
CLK
DI(IO0)
C107
100nF
25V,X7R
U16
W25Q256JVEIQ
Flash
D11
LRB520S-30T1G
R51
10K
1%
TP20
C112
1uF
10V,X5R
R89
10K
1%
TP10
C113
12pF
50V,NPO
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Sheet: /ESP32/
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Title:
Size: A3
Date:
Rev:
KiCad E.D.A. 9.0.9
Id: 5/9


U2
XU316-1024-QF60B-C24
+1V8
+3V3
VDDIO
VDD
GND
USB_VDD33
USB_VDD18
VDDIOB18
VDDIOB18
VDDIOL
VDDIOR
VDDIOT
PLL_AVDD
USB_DP
USB_DM
TCK
TMS
TDI
TDO
RST_N
NC
XIN
XOUT
X1D00
X1D01
X1D09
X1D10
X1D11
X1D13
X1D16
X1D17
X1D18
X1D19
X1D22
X1D34
SPI_CS_N
QSPI_CS_N
QSPI_D0
QSPI_D1
QSPI_D2
QSPI_D3
QSPI_CLK
SPI_CLK
I2S2_DATA
I2C2_SCL
I2C2_SDA
XOD31
XOD32
XOD33
RST_DAC
ESP_IO41
I2S2_BCLK
I2S2_MCLK
I2C_SCL
I2C_SDA
I2S2_LRCLK
XMOS_MUTE_DET
BUT_A
XOD42
XU316_WS2812
XOD43
ESP_IO40
I2S_DOUT
I2S_LRCK
I2S_BCLK
I2S_MCLK
VBUS_DETEC
MIC_DATA
XL_DN1
XL_DN0
XL_UP0
XL_UP1
MIC_CLK
I2S_DIN
I2S_DIN_SEC
I2S_LRCK
I2S_BCLK
I2S_MCLK
I2S_DIN_ESP
XMOS_USB_P
XMOS_USB_N
BOOT_SEL
FB6
600R
500mA
C41
1uF
10V,X5R
X1
24MHz 12pF
10ppm
C3
20pF
50V,NPO
C2
20pF
50V,NPO
R3
1M
R4
680R
5%
R59
100K
1%
R58
47K
5%
Q10
LBSS138WT1G
200mA
VBUS
VBUS_DETEC
C52
C53
C54
100nF
25V,X7R
FB9
120R
2A
C55
C56
C57
C58
C127
C128
C129
12pF
50V,NPO
Flash
U3
ZB25VQ32DSJG
/CS
DO(IO1)
/WP(IO2)
GND
VCC
/HOLD(IO3)
CLK
DI(IO0)
C1
100nF
25V,X7R
Mute Switch
SW1
SK-22H07
CH1A
CH1B
CH2A
CH2B
COM1
COM2
S1
S2
Mute_EN
R66
100R
R10
1K
C9
100nF
25V,X7R
U14
SN74LVC1G17DCKR
C60
100nF
25V,X7R
MUTE_ON
TP25
TP28
R114
0R
MUTE_ON
MUTE
0(Default)
OFF
1
ON
MIC Data Ctrl
U6
SN74LVC125APWR
1OE
1A
1Y
2OE
2A
2Y
3OE
3A
3Y
MIC_CLK
MIC_DATA_M
MIC_DATA
R8
100R
R7
47K
C137
12pF
50V,NPO
C6
100nF
25V,X7R
ESP_3V3
R98
10K
1%
R53
10K
1%
D13
D20
LRB520S-30T1G
R138
R139
0R
MIC Power
U18
SGM2036S-3.3XXDH4G/TR
IN
OUT
EN
GND
C11
2.2uF
10V,X7R
C10
100nF
25V,X7R
R12
3.3K
1%
R28
470K
1%
Q4
S9013W
25V,500mA
C103
10nF
50V,X7R
C110
1uF
10V,X5R
C12
100nF
25V,X7R
TP26
VDD_MIC
R11
47K
5%
TP13
MIC
U4
U5
MSM261DHP006
CLK
DATA
GND1
GND2
VDD
L/R
GND4
GND3
L
R
MIC_CLK_M
MIC_DATA_M
R109
R6
R110
100R
C133
C134
C135
C136
D1
D4
D10
D18
D19
D12
ESDPSA0402V05
C4
C5
100nF
25V,X7R
The distance between two microphones is 71mm.
DEBUG
R31
10K
1%
RST_N
TP46
R63
100R
XU316_RST
R29
470K
Q3
LBSS138WT1G
200mA
C92
100nF
25V,X7R
J5
C142
C143
TMS
TCK
TDO
TDI
RST_N
XL_DN1
XL_DN0
XL_UP0
XL_UP1
XMOS RST must after VDD and VDDIOB_18 Power Good.
0.9V
FB7
600R
500mA
C44
1uF
10V,X5R
C43
C45
C46
C47
C48
C49
C50
C51
100nF
25V,X7R
TP12
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Sheet: /XMOS/
File: XMOS.kicad_sch
Title:
Size: A3
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Rev:
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Id: 6/9



MIC	Q2_D	AMP_CTL
1	1	0
0	0	1

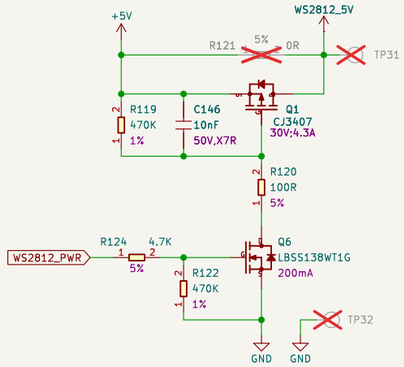
AMP_CTRL	AMP Status
0-0.5V	Shutdown
1.55V-	Class AB

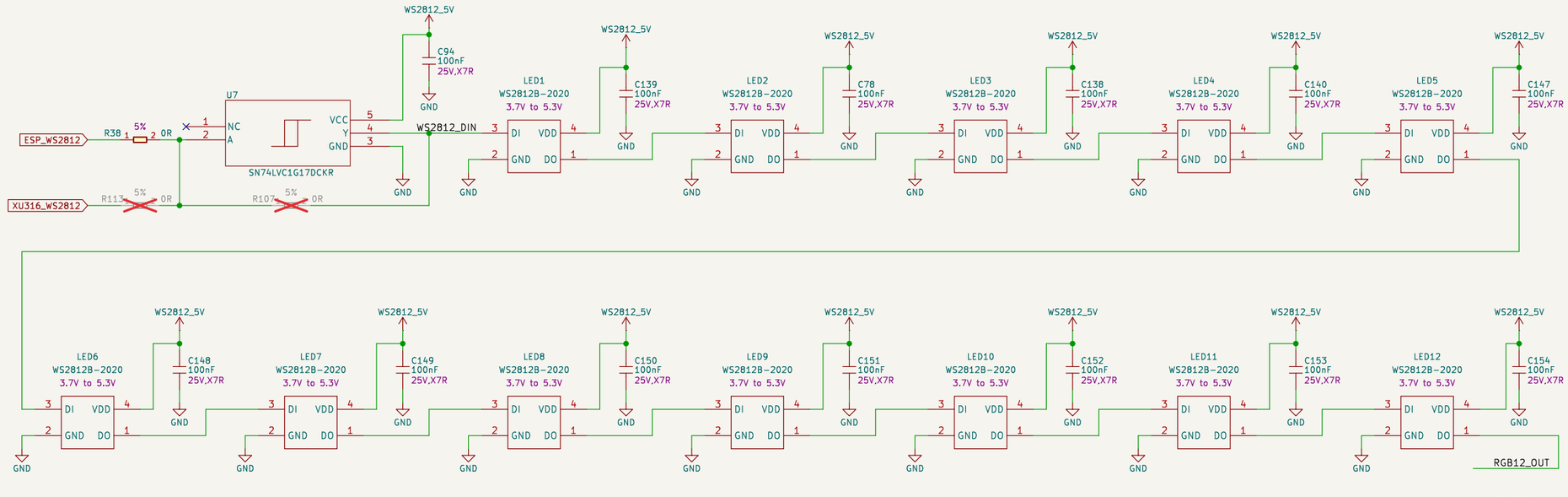
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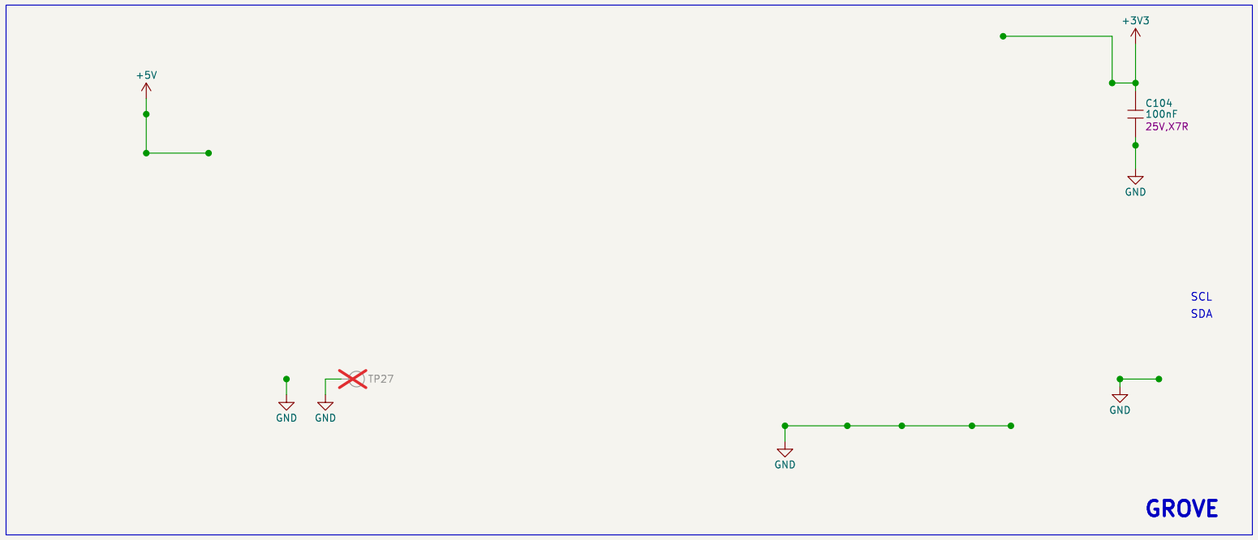
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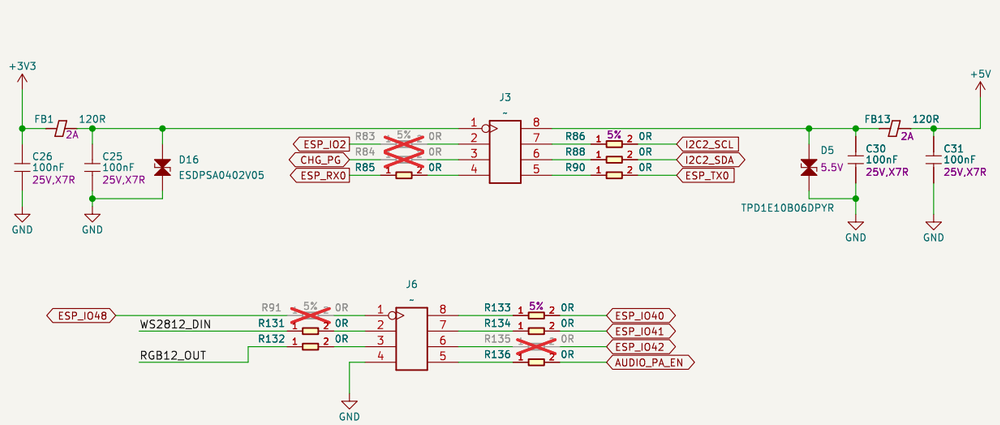
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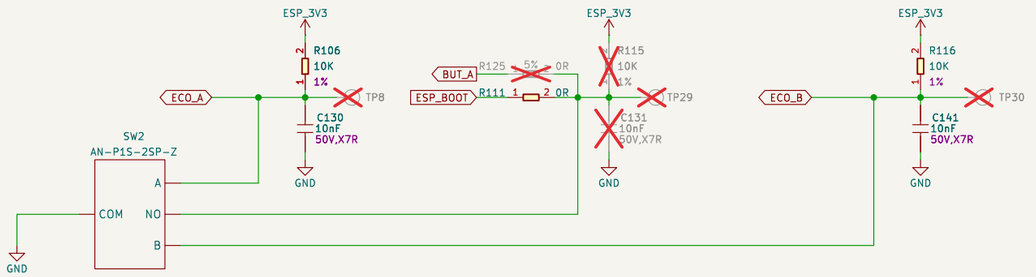
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KiCad E.D.A. 9.0.9		Id: 7/9







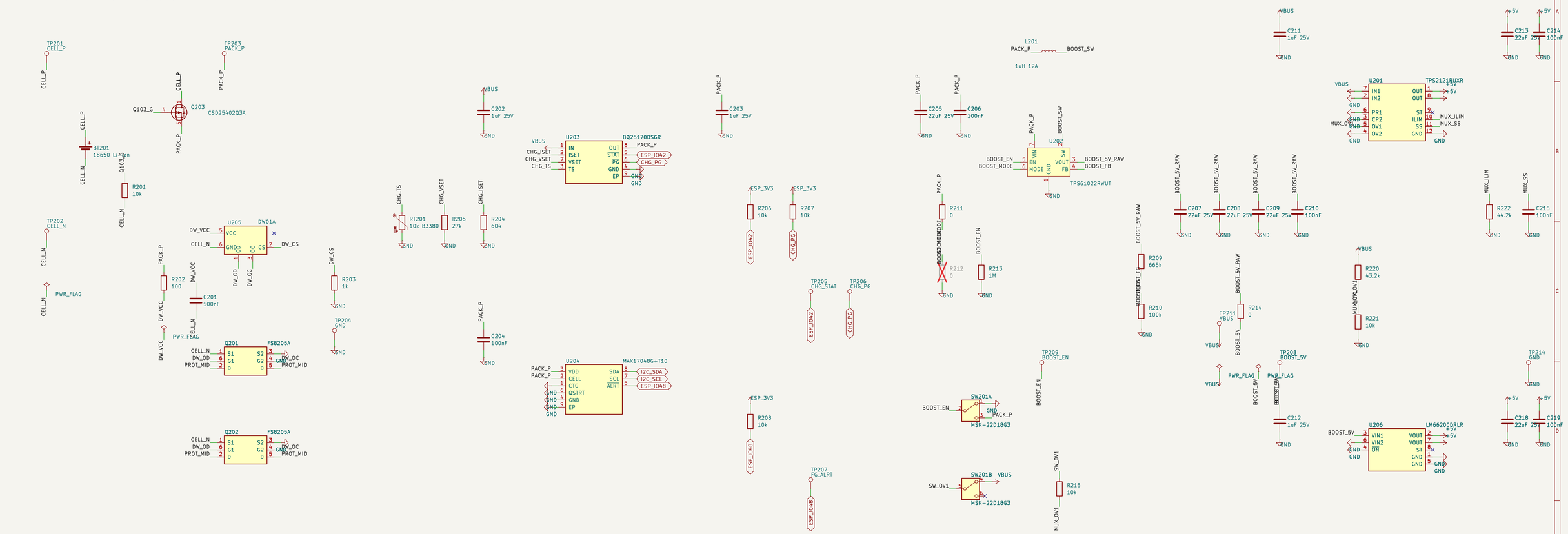




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Sheet: /LED & Connector/		
File: LED & Connector.kicad_sch		
Title:		
Size: A3	Date:	**Rev:**
KiCad E.D.A. 9.0.9		Id: 8/9

NEO v2 — bloque nuevo (diseño propio). Cálculos: tools/calc_power_v2.py; decisiones: docs/fase2-propuesta-bloques-nuevos.md



PROTECCIÓN DE CELDA (fase 2 §4.4)
Q203: inversión de polaridad. Puerta al borne − de la celda: VGS = −VBAT con la
celda bien puesta; con la celda invertida VGS > 0 y bloquea antes que nada.
DW01A: VOC 4,30 V, VOD 2,50 V, VEDI 150 mV (130–170).
2× FS8205A en paralelo: Rds serie 2×10,5 mΩ → disparo 4,06 A mín. (3,04 A pico).
R202/C201: filtro de VCC del DW01A. R203: resistencia de CS (hoja PUOLOP).

CARGADOR BQ25170 (SLUSDJ8A, fase 2 §4.3)
ICHG = KISET/RISET = 300 A·Ohm / 604 Ohm = 497 mA (447–546 mA). Precarga 20 %, fin 10 %.
RVSET 27 k → ≈ 4,20 V. NTC 10 k B3380: 0 °C = 1,073 V (VCOLD 1,04 V); 45 °C = 0,186 V (VHOT 0,188 V).
STAT → GPIO42 (ESP_IO42), /PG → GPIO1 (CHG_PG). Sin VIN-DPM: ver histéresis del multiplexor.

MEDIDOR MAX17048: VDD y CELL a PACK+, CTG y QSTRT a GND, ALRT → GPIO48 (ESP_IO48).
Dirección 0x36: sin colisión con AIC3204 0x18, OLED 0x3C, XU316 0x42.

ELEVADOR TPS61022 (SLVSDX7D, fase 2 §4.2)
VOUT = 0,6 V × (1 + 665k/100k) = 4,59 V (fase 4: por debajo de USB para el diodo ideal U206). MODE alto = PWM forzado (R212 0 Ohm; R213 1 MΩ).
L201 1 uH: IL pico 5,26 A a 2,7 V y 2 A < Isat 12 A; valle 3,44 A < ILIM mín. 6,5 A.
COUT 3×22 uF (~30 uF efectivos): eq. 8 pide 21,6 uF para 50 mV pp.
R214: puente 1206 de 0 Ohm: huella prevista para ferrita si la medida lo pide.

INTERRUPTOR SW201 (señal): OFF = EN a GND y OV1 forzado desde VBUS (corta USB y batería);
el cargador sigue cargando. ON = EN a PACK+. R213 fija EN durante el recorrido.

UNIÓN DE FUENTES (fase 4, docs/fase4-colocacion.md §2). Sustituye al multiplexor único de la fase 2:
R210 0 Ohm (R213 1 MΩ) y R211 (no cabe una pista de 1,5–2 A en la placa heredada).
U201 abajo: VBUS → +5V, ILIM 44,2 k (2,0–3,0 A), OV1 43,2k/10k (5,64 V), SS 100 nF. Entrada única.
U206 arriba: BOOST_5V (4,59 V) → +5V. Con USB, +5V (~4,8–5,0 V) > BOOST_5V + 70 mV: bloquea.
Sin USB conduce sin umbral ni histéresis. El plano +5V de In1 une ambos extremos.



Bloque nuevo de NEO v2 (no forma parte del diseño de Nabu Casa)
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NEO v2 — techmiguel

Sheet: /Alimentacion/
File: Alimentacion.kicad_sch

Title: NEO v2 — Alimentación: batería, carga, elevador y multiplexor

Size: A2	Date:	**Rev:**
KiCad E.D.A. 9.0.9		Id: 8/9

NEO v2 — bloque nuevo (diseño propio)

PWR_FLAG
+3V3
TP215
OLED_3V3
FB201
120R@100MHz
C216
10uF 10V
C217
100nF
J201
OLED I2C
R225
22
R226
22
D201
TPD1E04U04
D202
TPD1E04U04
R227
4.7k
R228
4.7k
TP216
I2C_SDA
TP217
I2C_SCL
TP218
I2C2_SDA
TP219
I2C2_SCL
TP220
I2S_MCLK
TP222
I2S_LRCK
TP223
I2S_DIN_SEC
TP224
I2S_DIN_ESP
TP225
I2S2_BCLK
TP226
I2S2_LRCLK
TP227
I2S2_DATA
TP228
XU316_RST
TP229
ESP_IO40
TP230
ESP_IO41
TP231
ESP_IO2
TP232
ESP_IO46
TP233
GND
TP234
GND
TP235
GND

CONECTOR OLED (fase 2 §4.7)
Pines en el orden más común de los módulos SSD1306: GND, VCC, SCL, SDA.
R227/R228: segundo par de pull-ups. Con el MAX17048 y 10 cm de cable, el bus llega a ~112 pF: 4,7 k solos dan tr = 446 ns (> 300 ns); con 2,35 k, 223 ns.
D201/D202: 0,5 pF para no consumir el margen de capacidad.

PUNTOS DE PRUEBA (fase 2 §4.8): pad SMD Ø1,0 mm sobre la pista o con tramo corto (fase 4).
I2S: ~0,3 pF por pad y < 1 pF por tramo, despreciable frente a los ~24,576 MHz.
I2S_BCLK no tiene PT: su pista corre en el haz del bus, sin hueco para UCEPP en ninguna cara (fase 4); se mide en el pad 2 de R71 o en el pad 1 de C128 (cara B).

NEO v2 — techmiguel		
Sheet: /Expansion/		
File: Expansion.kicad_sch		
Title: NEO v2 — Conector OLED, pull-ups I2C y puntos de prueba		
Size: A3	Date:	Rev:
KiCad E.D.A. 9.0.9		Id: 9/9